

Lecture 7 2026-09-22

Last time: equality-constrained Newton

Today: N.C.O. for inequality constraints

Review N.C.O. we've seen so far:

$$\min_x f(x)$$

$$f \in \mathcal{C}^2$$

sub. to $h(x)=0$

$$h \in \mathbb{C}^2$$

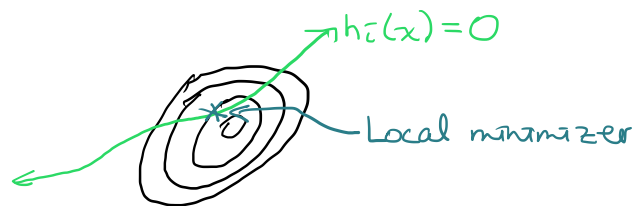
Unconstrained case: $\min_x f(x)$

Draw level curves of $f(x)$



$$\text{N.C.O.}: \nabla_x f(x^*) = 0$$

Equality-constrained case : $\min_x f(x)$
sub. to: $h(x)=0$



$$\text{N.C.O.}: \nabla_x f(x^*) + y_i^* \nabla_x h(x^*) = 0$$
$$h(x^*) = 0$$

Once we have multiple constraints:

$$\mathcal{L}(x, y) = f(x) + y^T h(x)$$

$$\rightarrow \begin{cases} \nabla_x \mathcal{L}(x^*, y^*) = 0 \\ \nabla_y \mathcal{L}(x^*, y^*) = 0 \end{cases} \left. \begin{array}{l} \text{Introduced residual} \\ r(x, y) \end{array} \right\}$$

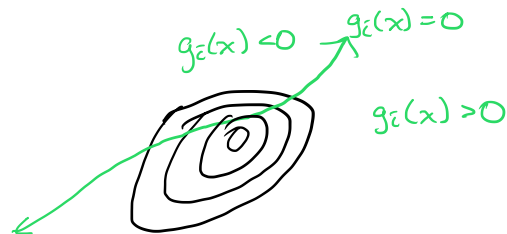
$\Rightarrow$ Solving for (x^*, y^*) becomes a root finding problem $r(x, y) = 0$

Today:

Inequality-constrained optimization:

$$\begin{aligned} \min_x & f(x) \\ \text{subj. to: } & g(x) \leq 0 \end{aligned}$$

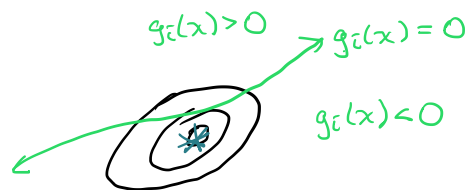
What are N.C.O. with inequality constraints?



We'll consider both cases for how x^* satisfies $g(x^*) \leq 0$

1. "Inactive" constraint

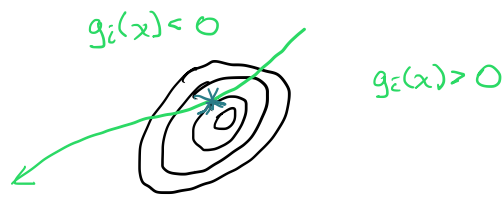
$\rightarrow g(x^*) \leq 0$ doesn't "influence" problem



If $g_i(x^*) < 0 \rightarrow$ N.C.O.: $\nabla_x f(x^*) = 0$

2. "Active" constraint

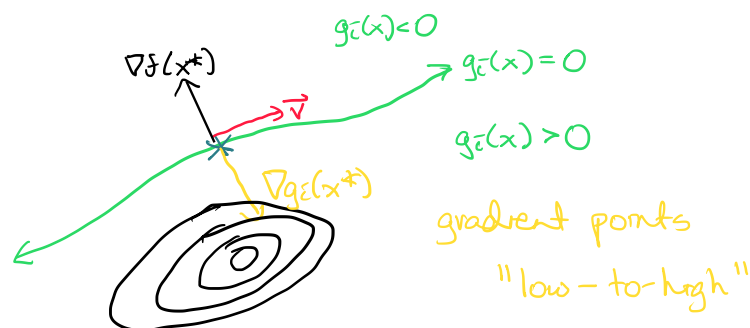
$$g_i(x) < 0 \rightarrow g_i(x) = 0$$



Using geometric arguments from last time for $h_i(x^*) = 0$,

$$\text{if } g_i(x^*) = 0 \rightarrow \text{N.C.O.: } \nabla_x f(x^*) + z_i^* \nabla_x g_i(x^*) = 0$$

Note: this time, we have a specific direction for z_i



With equality constraints, we projected $\nabla f(x)$ along local orthogonal $\nabla_x g_i(x)$ and local tangential $\vec{v}$

$$\nabla_x f(x) = z_i \nabla g_i(x) + \beta \vec{v}$$

we argued, stopping condition is when $\beta \vec{v} = 0$

This time we have that $\nabla_x f(x)$ and $\nabla_x g_i(x)$

must be **anti-parallel** at x^*

$$\Rightarrow \nabla_x f(x^*) = -z_i \nabla_x g_i(x^*)$$

$$z_i > 0$$

$$\rightarrow \text{N.C.O.: } \nabla f(x^*) + z_i^* \nabla g_i(x^*) = 0$$

⇒ Putting the inactive and active cases together,
the "pseudo code" would look like:

```
if  $g_i(x^*) = 0$                                      // Active case
    NCO:  $\nabla_x f(x^*) + z_i^* \nabla_x g_i(x^*) = 0$ 
         $z_i^* > 0$ 
else
    // Inactive case
    NCO:  $\nabla_x f(x^*) = 0$ 
```

We can't differentiate through an if/else conditional →
how can we rewrite this?

Complementarity constraints: $z_i^* \cdot g_i(x^*) = 0$
 $z_i^* \geq 0$

This allows us to impose a single set of NCO
and allow for existence of solutions in both active
and inactive cases.

⇒ Putting into Lagrangian form:

$$\mathcal{L}(x, z) = f(x) + \sum_{i=1}^p z_i g_i(x)$$

NCO:

$$1. \nabla_x \mathcal{L}(x^*, z^*) = \nabla_x f(x^*) + \sum_{i=1}^p z_i^* \nabla_x g_i(x^*) = 0$$

$$2. \quad g_i(x^*) \leq 0 \quad i=1, \dots, p$$

$$3. \quad z_i^* \cdot g_i(x^*) = 0 \quad i=1, \dots, p$$

$$4. \quad z_i^* \geq 0 \quad i=1, \dots, p$$

These are new as compared
with equality constrained NCO

How does this recover both active and inactive cases?

1. Inactive case:

$$\text{If } g_i(x^*) < 0, \text{ from \#3: } z_i^* \cdot g_i(x^*) = 0 \Rightarrow z_i^* = 0$$

$$\text{From \#1, if } z_i^* = 0 \Rightarrow \nabla_x \mathcal{L}(x^*, z^*) = \nabla f(x^*) = 0$$

3. Active case:

$$\text{If } g_i(x^*) = 0, \text{ then \#3 allows } z_i^* \geq 0$$

$$\text{From \#1, we have } \nabla_x \mathcal{L}(x^*, z^*) = \nabla_x f(x^*) + z_i^* \nabla_x g_i(x^*) = 0$$

Let's add back m equality constraints:

$$\min_x f(x)$$

$$f, h_i, g_i \in \mathcal{C}^2$$

$$\text{subj. to: } h_i(x) = 0 \quad i=1, \dots, m$$

$$g_i(x) \leq 0 \quad i=1, \dots, p$$

$$\mathcal{L}(x, y, z) = f(x) + \underbrace{\sum_{i=1}^m y_i h_i(x)}_{\text{From equality constraints}} + \underbrace{\sum_{i=1}^p z_i g_i(x)}_{\text{From inequality constraints}}$$

Karush Kuhn Tucker (KKT) conditions:

$$1. \nabla_x \mathcal{L}(x^*, y^*, z^*) = 0$$

$$= \nabla_x f(x^*) + \sum_{i=1}^m y_i^* \nabla_x h_i(x^*) + \sum_{i=1}^p z_i^* \nabla_x g_i(x^*) = 0$$

$$2. \nabla_y \mathcal{L}(x^*, y^*, z^*) = 0$$

$$\Rightarrow h_i(x^*) = 0 \quad i = 1, \dots, m$$

$$3. g_i(x^*) \leq 0 \quad i = 1, \dots, p$$

$$4. z_i^* \cdot g_i(x^*) = 0 \quad i = 1, \dots, p$$

$$5. z_i^* \geq 0 \quad i = 1, \dots, p$$

